

Problem 1. Airport data

**(a) What is the  $\alpha$  if the distribution is power law?**

Problem 1. Airport data

**(b) What is the  $\lambda$  if the distribution is exponential?**

Problem 1. Airport data

(c) **What is  $[a, b]$  if the distribution is uniform?**

Problem 1. Airport data

(d) **What is  $\mu$  and  $\sigma$  if the distribution is normal?**

Problem 1. Airport data

(e) Screen shot of the visualization of distribution of data

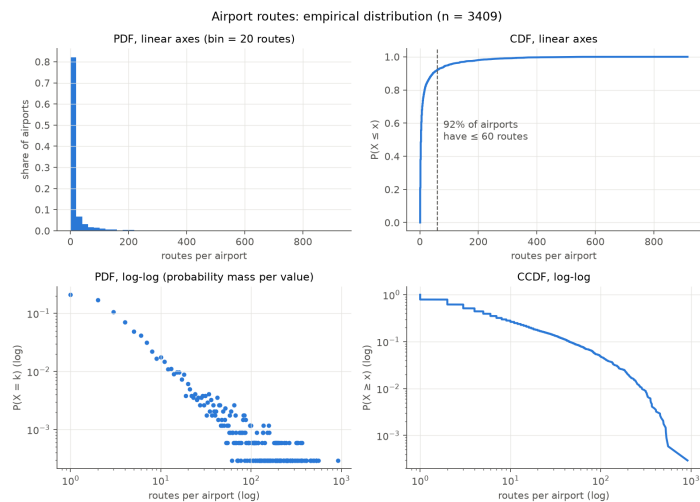


Figure 1: Distribution of the airport routes data.

Problem 1. Airport data

(f) Screen shot of the visualization of distribution of data if it was power law

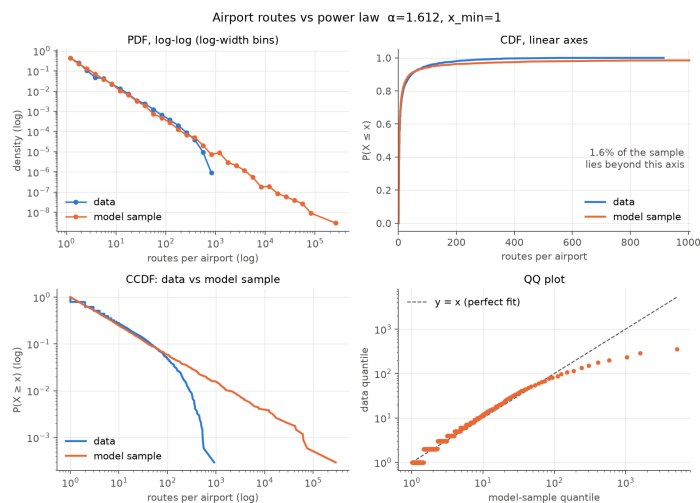


Figure 2: Airport data vs. samples from the fitted power law model.

Problem 1. Airport data

(g) Screen shot of the visualization of distribution of data if it was exponential

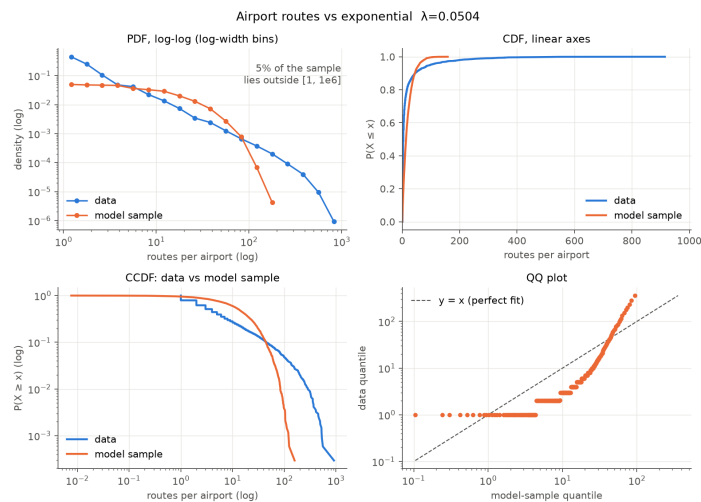


Figure 3: Airport data vs. samples from the fitted exponential model.

Problem 1. Airport data

(h) Screen shot of the visualization of distribution of data if it was uniform

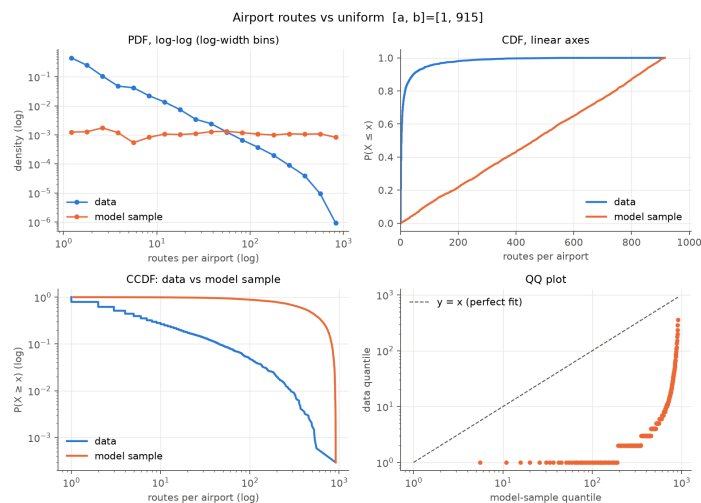


Figure 4: Airport data vs. samples from the fitted uniform model.

Problem 1. Airport data

(i) Screen shot of the visualization of distribution of data if it was normal

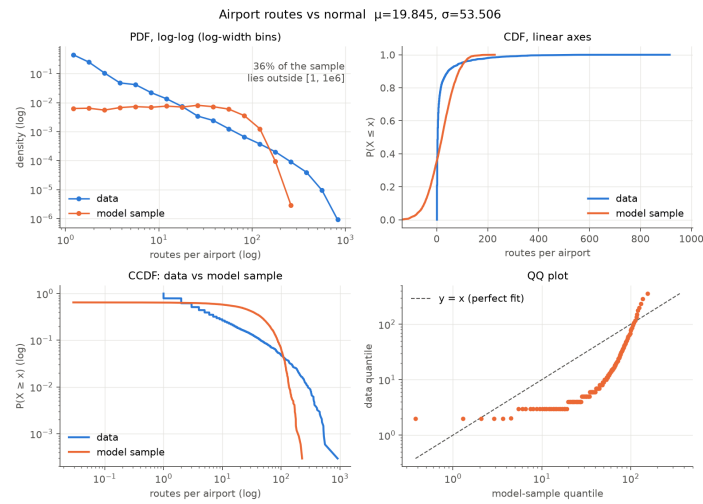


Figure 5: Airport data vs. samples from the fitted normal model.

Problem 1. Airport data

(j) What is the distribution of data based on your observation? Discuss

Problem 1. Movie Rating data

**(a) What is the  $\alpha$  if the distribution is power law?**

Problem 1. Movie Rating data

**(b) What is the  $\lambda$  if the distribution is exponential?**

Problem 1. Movie Rating data

**(c) What is  $[a, b]$  if the distribution is uniform?**

Problem 1. Movie Rating data

**(d) What is  $\mu$  and  $\sigma$  if the distribution is normal?**

Problem 1. Movie Rating data

(e) Screen shot of the visualization of distribution of data

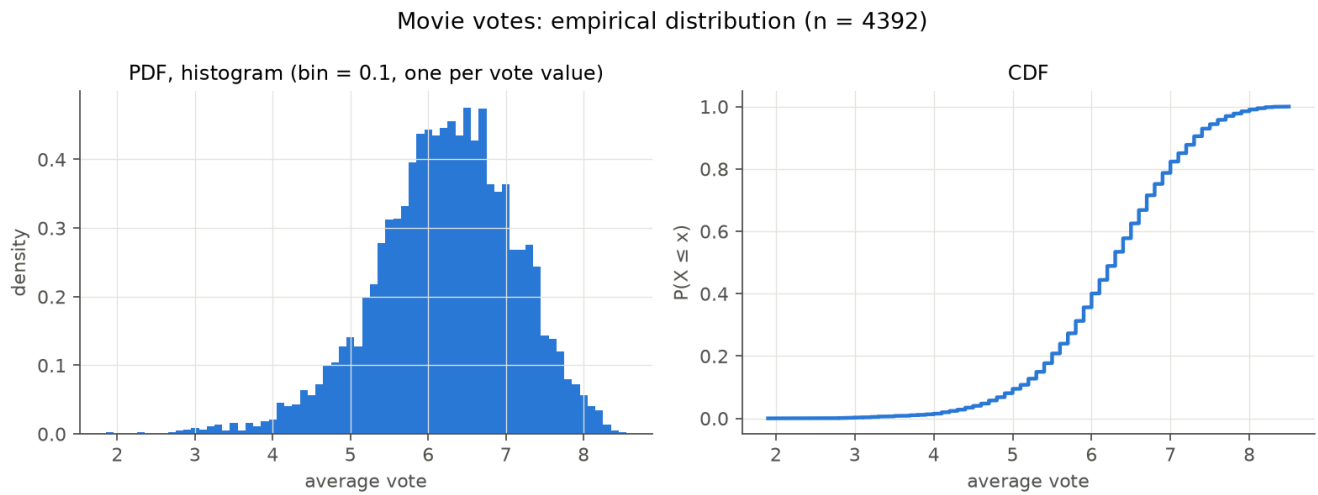


Figure 6: Distribution of the movie rating data.

Problem 1. Movie Rating data

(f) Screen shot of the visualization of distribution of data if it was power law

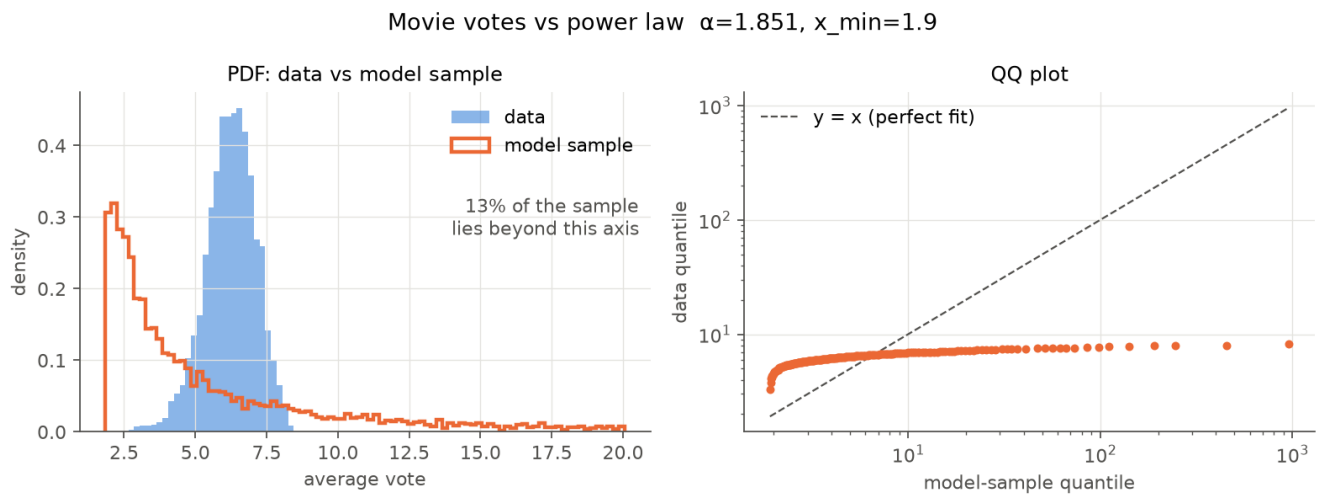
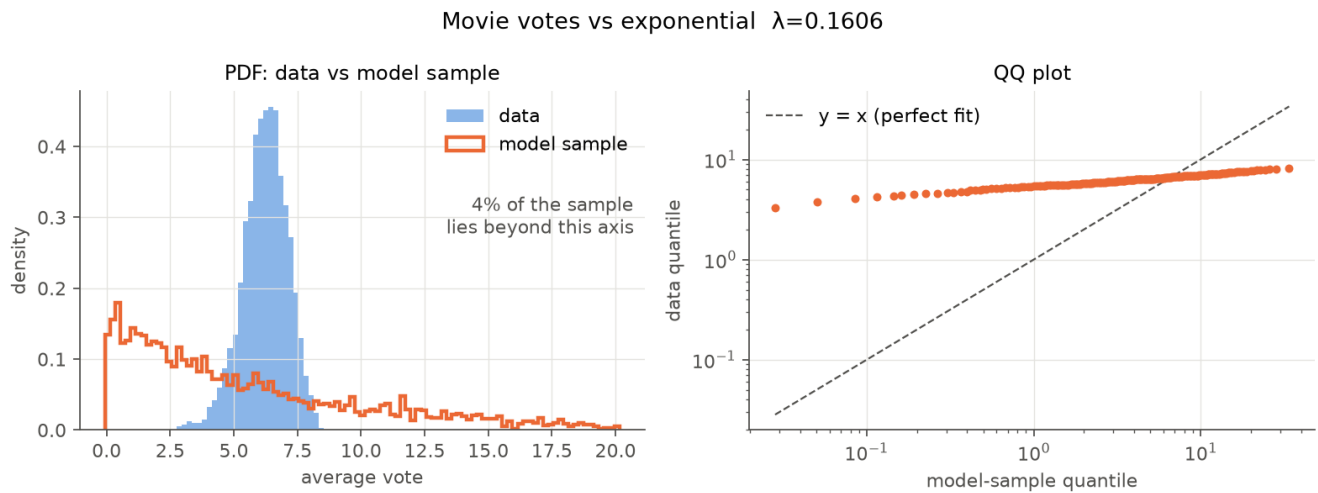


Figure 7: Movie rating data vs. samples from the fitted power law model.

Problem 1. Movie Rating data

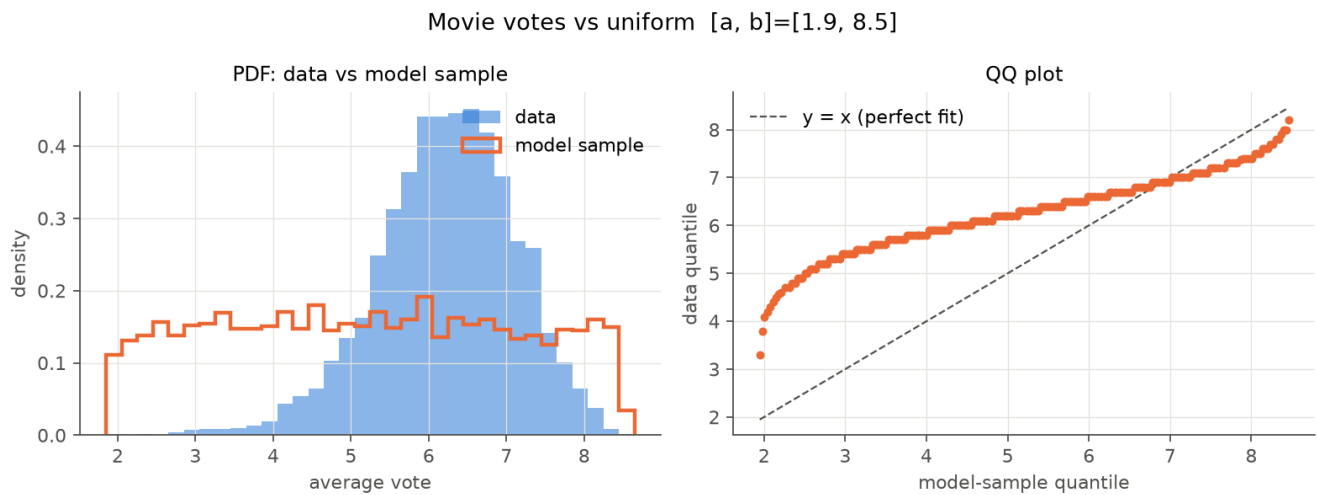
(g) Screen shot of the visualization of distribution of data if it was exponential



**Figure 8:** Movie rating data vs. samples from the fitted exponential model.

Problem 1. Movie Rating data

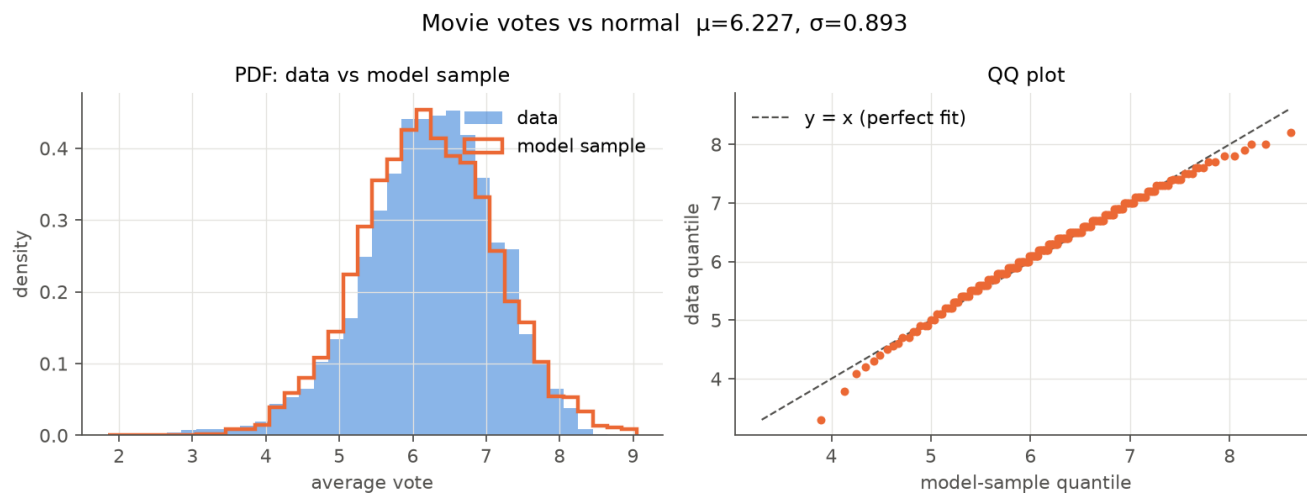
(h) Screen shot of the visualization of distribution of data if it was uniform



**Figure 9:** Movie rating data vs. samples from the fitted uniform model.

Problem 1. Movie Rating data

(i) Screen shot of the visualization of distribution of data if it was normal



**Figure 10:** Movie rating data vs. samples from the fitted normal model.

Problem 1. Movie Rating data

(j) What is the distribution of data based on your observation? Discuss

Problem 2. Age data

(a) Screenshot of distribution/histogram of the age

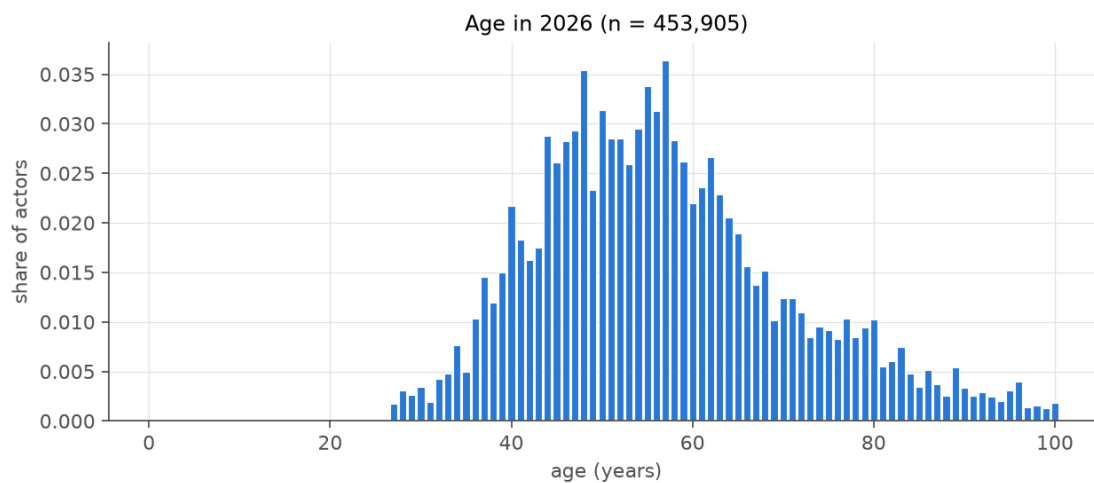


Figure 11: Distribution of age.

Problem 2. Age data

(b) Screenshot of distribution/histogram of the first digit of age

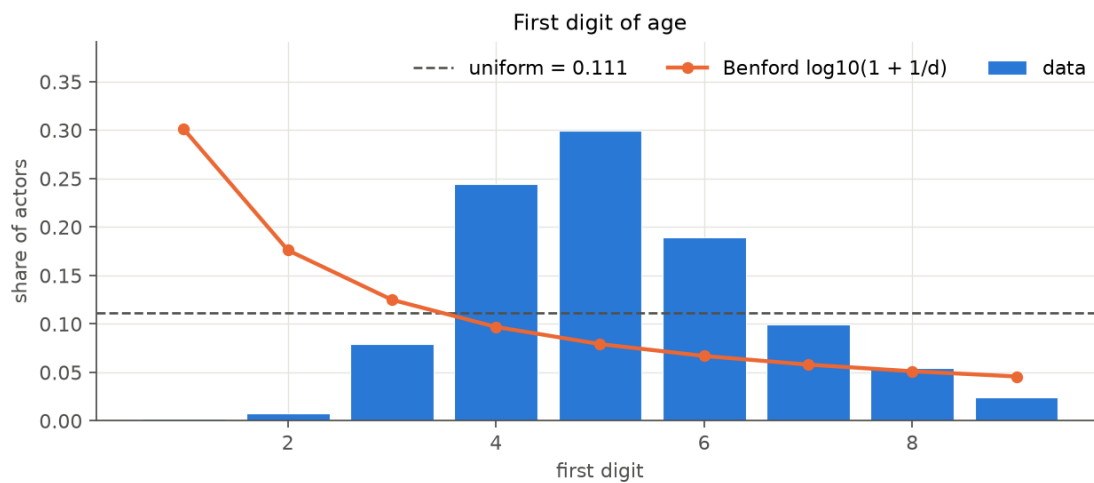
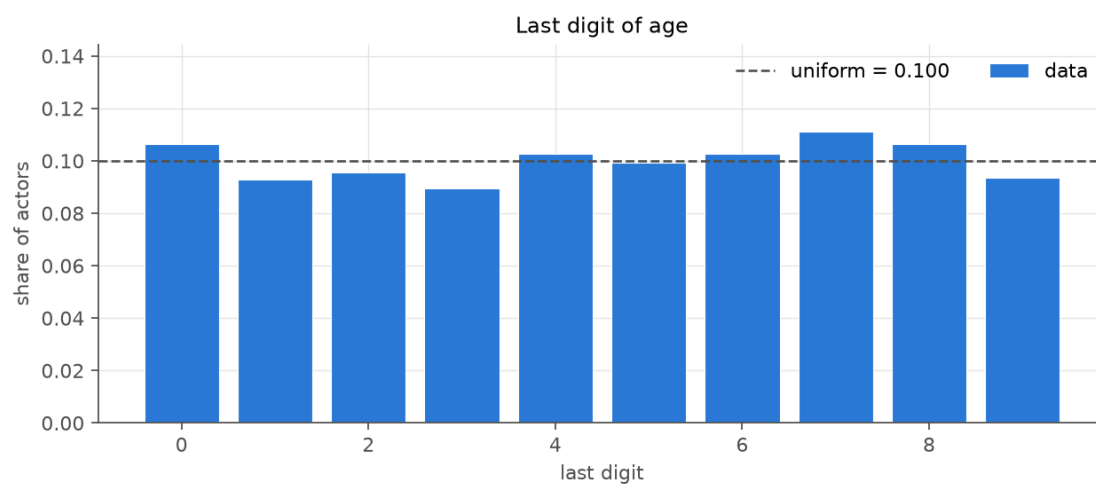


Figure 12: Distribution of the first digit of age.

Problem 2. Age data

(c) Screenshot of distribution/histogram of the last digit of age



**Figure 13:** Distribution of the last digit of age.

Problem 2. Age data

(d) Discussion (what are those distribution?) Why do you think it is like this?